Graded Lab 1

ERD and Relational Mapping

E-Commerce Platform

Q1. ERD Design

Design a complete ER diagram based on the following progressively expanding requirements:

Phase 1

- A user has a unique ID, name, and email
- A user can place multiple orders
- Each order has a unique ID and order date
- Each order is placed by exactly one user

Phase 2

- Each order contains one or more products
- Each product has a unique ID, name, and price
- A product can appear in multiple orders

Phase 3

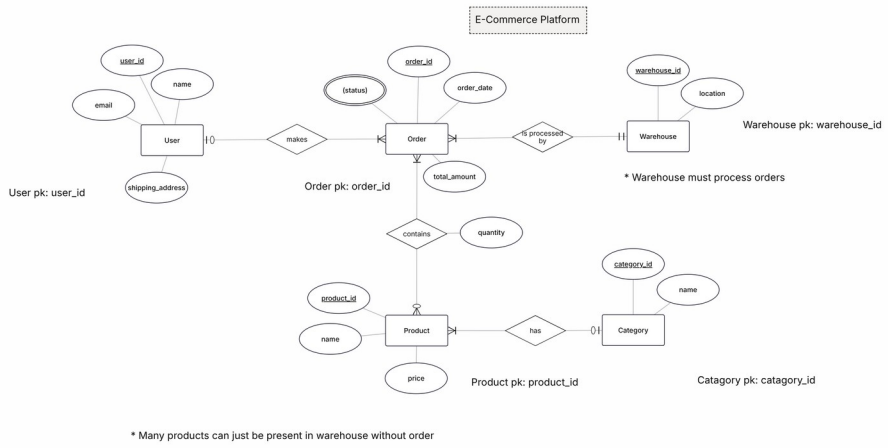
- Each product belongs to one category
- A category has a unique ID and name
- Each order is processed by one warehouse
- A warehouse has a unique ID and location
- A warehouse can process multiple orders

Phase 4

- Store quantity of each product in an order
- Each order has status: pending, shipped, delivered
- Store total amount for each order
- Each user has a shipping address

Instructions:

- Draw a complete ERD after incorporating all phases
- Clearly indicate Primary Keys
- Clearly indicate Cardinality (1:1, 1:N, M:N)
- Mention assumptions (if needed)

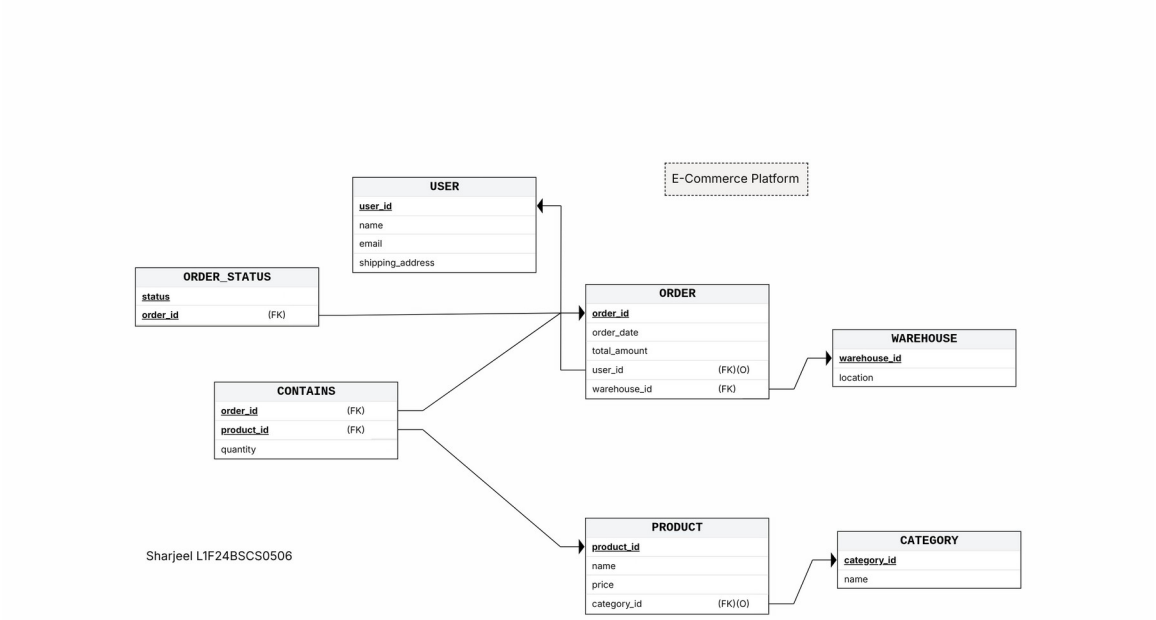


Q2. Relational Mapping (Schema Design)

Convert your final ERD into a Relational Schema.

Requirements:

- Clearly define all relations (tables)
- Specify Primary Keys
  User: user_id
  Order: order_id
  Product: product_id
  Warehouse: warehouse_id
  Catagory: catagory_id

- Specify Foreign Keys

Order: user_id, warehouse_id
Product: catagory_id
Contains: order_id, product_id acting as both fk and pk